EAST WEST UNIVERSITY

Project Report
On

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The Departmental Store Management System

Project Overview

The Departmental Store System is a Java-based application designed to manage products, employees, and billing operations. Using object-oriented principles, the system features classes such as **Product, Invoice, Employee, StoreEmployee, Manager, and Main**. It demonstrates inheritance, abstraction, polymorphism, file handling, exception handling, and console-based user interaction.

Objectives

- Apply Object-Oriented Programming principles in a practical project.
- Implement file handling for employee persistence.
- Build a modular, maintainable, and reusable codebase.
- Handle exceptions and provide robust console interactions.
- Demonstrate inheritance, abstraction, and polymorphism in real-world scenarios.

System Features

Feature	Handled By
View Products	Product
Create Invoice	Invoice, Product, Main
View Employees	StoreEmployee, Employee
Add/Remove Employee	Manager, StoreEmployee, Employee
Manager Login	Manager

Java Features

- **Classes & Objects:** Core classes like Product, Invoice, Employee, StoreEmployee, Manager, Main.
- **Abstraction:** Employee is abstract.
- **Polymorphism:** Overridden methods (e.g., `displayInfo()`).
- **Encapsulation:** Private/protected fields with getters/setters.
- **Association:** Manager manages StoreEmployee objects; Invoice contains Product objects.
- **Generics:** Used in Product and Invoice for price and product lists.
- **File Handling:** Employee data stored in text files.
- **Exception Handling:** InputMismatchException and IOException handled.
- **Console I/O:** Scanner input and formatted output for receipts.
- **Inheritance:** StoreEmployee and Manager extend Employee.

Class Structure

Employee <<abstract>>

- name: String

- id: int

- position: String

+ displayInfo()

+ getName(), getId(), getPosition()

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StoreEmployee

- salary: double

+ displayInfo(), getSalary(), setSalary()

+ showAllEmployees()

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Manager

- employees: List<StoreEmployee>

+ addEmployee(), removeEmployee()

+ login(), showEmployeeManagementMenu()

Product<T>

- name, code, price

+ displayInfo(), getAllProducts(), getProductByCode(), displayAllProducts()

Invoice

- items, totalAmount, dateTime

+ addItem(), getTotalAmount(), generateReceipt()

Main

+ main(), createInvoice()

Data Handling

- Employee info saved in **employees.txt**.
- Default employees added if file missing.
- Invoice data processed during runtime and displayed as receipts.
- Ensures consistency of employee records between sessions.

Conclusion

This project demonstrates the effective use of Java programming and core object oriented principles in developing a functional Departmental Store Management System. It integrates key features like data encapsulation, inheritance, abstraction, and file handling to simulate a real-world academic environment. The codebase follows a modular and extensible design, allowing easy maintenance and future enhancements such as GUI integration, database connectivity, and user authentication. Overall, it serves as a solid foundation for understanding howsoftware systems are designed and implemented using Java.